MACHINE LEARNING PROJECT · CLASS PRESENTATION

CrowdSense AI

Do temporal crowd-movement features improve anomaly prediction over static crowd-density features alone?

Detection · tracking · optical flow · window features · sequence-held-out evaluation

UCSD Anomaly Detection Dataset v1p2 · CPU-only · seed 42

Two ways to read a crowd

Static density

How many people are there, and how are they arranged?

Person count, density per megapixel, grid concentration, spread in x and y, grid entropy — six features computed from one frame.

Temporal movement

How is the crowd moving?

Speed, acceleration, direction changes, flow, abrupt stops, reversals, track persistence — seventeen features computed over a 30-frame window.

Research question: do movement features improve anomaly prediction over static density alone?

Pre-declared success: held-out test AUC ≥ 0.75 with better F1. The honest answer is on slide 7.

02 · DATA

UCSD Anomaly Detection v1p2, split by sequence

98	18,560	5,693	3,222
sequences (videos)	frames	anomalous-frame labels	30-frame windows

Grouped split by sequence id — no video spans two splits



Test set = 495 windows, of which 179 are anomalous.

- Splits enforced in code: assert_group_disjoint
- Window label = 1 if any frame is anomalous (frame-level ground truth)
- No counting dataset — counts are detector estimates, never MAE/RMSE

Why grouped splits?

Frames from one video are near-duplicates. A random frame split would leak them into test and inflate the score; grouping by sequence keeps test videos unseen.

How a window gets its label

30 frames, schematic



03 · METHOD

From video to one anomaly score per window

Video
MP4 / AVI / MOV or a photo

→

Detect
YOLOv8n persons conf 0.05 · 640 px

→

Track
ByteTrack IDs across frames

→

Flow
Farneback optical flow

→

Window
30 frames stride 5

→

Score
23 features → model → P(anomaly)

Two paths through the same pipeline

- Photo: one frame → detection + six static features (no anomaly score)
- Video: window → all 23 features → per-window anomaly score

Why a window?

Motion needs time. One frame has no speed, direction, or flow — so no anomaly score is shown for photos.

Detector calibration: a 40-frame grid chose conf 0.05 / 640 px by a pre-declared rule — most detections under a 2-hour rebuild budget (16.40 vs 10.25 boxes/frame).

04 · FEATURES

6 static + 17 movement = 23 features

Static — one frame

6 features

- person_count — detector estimate
- density_per_mpx — persons / megapixel
- grid_concentration — fraction 0–1
- spread_x — normalized
- spread_y — normalized
- grid_entropy — normalized 0–1

Movement — 30-frame window

17 features

- density_delta
- mean_speed_pxf
- speed_std_pxf
- direction_entropy
- directional_consistency
- flow_mean_magnitude
- flow_std_pxf
- track_count
- track_persistence
- nn_distance_mean
- nn_distance_std
- abrupt_stop_fraction
- direction_reversal_rate
- fast_mover_fraction
- speed_p90
- track_turnover_rate
- flow_direction_inconsistency

All speeds are px/frame at processed resolution — never m/s (no scene-scale calibration).

Decisions on train/val only; test touched once per config

1. **Tune** — GroupKFold on train. Five algorithms: logreg, decision tree, random forest, SVM, XGBoost.
2. **Threshold** — Pick the decision threshold on validation F1, never on test.
3. **Ensemble** — Soft-vote RF + SVM + XGB; members chosen by validation AUC.
4. **Test** — Four pre-declared configurations, each evaluated once on the held-out split.

Why the cap matters: every extra look at test inflates the score. The screen, tuning, and threshold all avoid test.

06 · RESULTS

Four pre-declared runs on the held-out test split

#	Configuration	Feat.	Thr.	Test AUC	Test F1
1	Baseline — original 15 features, logreg, untuned	15	0.50	0.6677	0.5068
2	+ schema-v2 movement features, logreg, untuned	23	0.50	0.6475	0.4892
3	**+ grouped-CV tuning + val threshold (RF)**	**23**	**0.25**	**0.6714**	**0.5311**
4	+ soft-vote ensemble RF + SVM + XGB	23	0.20	0.6705	0.5090

Source: reports/results/experiment_attribution.json · best row highlighted · threshold chosen on validation

0.6714
best test AUC (run 3)

0.5311
best test F1 (run 3)

−0.004
vs. previous AUC 0.675

07 · THE HONEST RESULT

The 0.75 AUC goal was not met — the ceiling held near 0.67

- Static density alone is near chance: AUC 0.43–0.56 across five algorithms.
- Movement carries the signal, but absolute performance plateaus near 0.67 on sequence-held-out UCSD.
- The calibrated detector raised detections from 10.25 to 16.40 boxes/frame and did not lift the old baseline.
- The eight new features hurt untuned; they only pay off with grouped-CV tuning and a validation threshold.
- The ensemble did not beat the tuned single model.

What did improve

Test F1: 0.517 → 0.5311 with the validation-tuned threshold.

Why the ceiling is real

Window labels inherit frame-level ground truth; normal crowds still contain unusual motion. With 15 test sequences, a 0.08 AUC gain is hard to establish.

AUC: achieved vs pre-declared goal



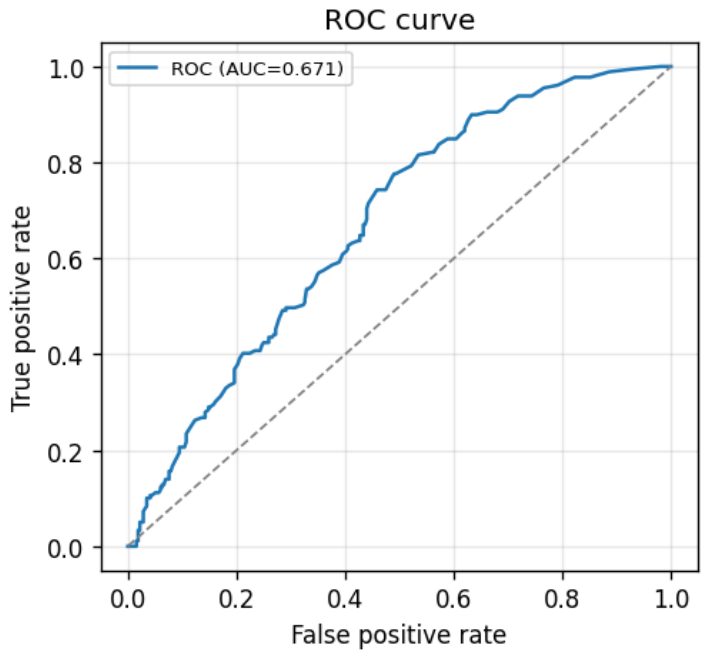
Reported as measured: the gain is in F1, not AUC. The earlier 0.675 baseline is not re-labelled, and no metric is dropped or rounded up.

Movement features beat static density for every algorithm

Algorithm	A static (6)	B movement (17)	C both (23)	C − A
logreg	0.505 / 0.159	0.653 / 0.490	0.648 / 0.489	+0.142
SVM	0.427 / 0.162	0.665 / 0.386	0.682 / 0.429	+0.255
XGBoost	0.499 / 0.256	0.629 / 0.420	0.614 / 0.351	+0.115
Random forest	0.470 / 0.267	0.635 / 0.385	0.671 / 0.376	+0.202
Decision tree	0.563 / 0.426	0.552 / 0.409	0.593 / 0.446	+0.029

Cells are test AUC / test F1 at threshold 0.5. C − A is the AUC gain from adding movement.

- C − A is positive for all five algorithms — the answer to the research question is yes.
- Static-only is near chance; movement-only already recovers most of the signal.



Random forest, config C (static + movement), test ROC — AUC 0.671.

09 · LIMITATIONS

What this work does not claim

No density CNN

CPU-only machine. Counting is a YOLOv8n detection baseline, not a density regressor.

No counting ground truth

QNRF and ShanghaiTech were dropped from scope, so no MAE or RMSE is reported anywhere.

No congestion labels

Experiment E (congestion classification) is hard-gated off as not supported.

Speeds are px/frame

No scene-scale calibration, so no m/s conversion is possible.

Frame-level labels

A window is anomalous if any frame in it is; the label is coarser than the score.

Calibration is a choice

conf 0.05 / 640 px maximises detections under a 2-hour budget — recall over precision.

Any claim about counting accuracy, m/s speeds, or congestion classes would be unsupported by this data.

The same pipeline behind a three-tool web app

Photo Analysis

Detect, annotate, 4×4 density grid, six static features.

Video Analysis

Windows, per-window anomaly score, top-scoring windows.

Live Camera

Browser camera → single-frame detection, no anomaly score.

Browser
React + TypeScript
(Vite SPA)

→

API
FastAPI
/api/*

→

Pipeline
detect → track →
flow → features

→

Model
bundle +
P(anomaly)

Demo

python -m uvicorn backend.app:app → http://127.0.0.1:8000 · missing metrics render as —, counts are labelled estimates, speeds stay px/frame

Three things to remember

1

Movement matters

Temporal features beat static density for every algorithm tested; static-only is near chance.

2

The ceiling is real

Sequence-held-out UCSD plateaus near 0.67 AUC. The defensible gain is F1, not AUC.

3

Honesty is a design choice

Grouped splits, four pre-declared runs, validation-only selection, and no invented metrics.

Next: more sequences for tighter intervals, a GPU density CNN, and richer anomaly labels. Questions?